Building and Environment 290 (2026) 114140

Contents lists available at ScienceDirect

Building and Environment

journal homepage: www.elsevier.com/locate/buildenv

The role of personality in multi-domain environmental perception: An occupant-centric approach

Agnese Chiucchiù [a], Veronica Martins Gnecco [b], Ilaria Pigliautile [a], Anna Laura Pisello [a,b,*]

[a] EAPLAB at CIRIAF Interuniversity research centre on pollution and environment Mauro Felli, University of Perugia, Italy
[b] Department of Engineering, University of Perugia, Italy

ARTICLE INFO

Keywords:
Multi-domain comfort
Human-centric design
Energy efficiency
Personality
Personalized comfort

ABSTRACT

The built environment accounts for nearly one-third of global energy demand, underscoring the need to enhance building performance through strategies that integrate both technological solutions and human factors. Occupants play a pivotal role in this process, as their comfort and well-being are shaped by complex interactions across multiple environmental domains. This study investigates cross-domain effects between thermal and acoustic environments, with a specific focus on how personality traits modulate perceptual responses. An experimental campaign was conducted in a controlled test room under two thermal conditions: a neutral phase (21 °C) and a mild discomfort phase (23 °C). In each phase, 25 participants evaluated seven soundscapes with distinct acoustic characteristics, ranging from natural to anthropogenic. Continuous monitoring of environmental parameters ensured accurate boundary conditions, while personality traits were assessed to capture inter-individual variability. Results show that even slight thermal variations influence acoustic perception, producing systematic negative shifts in the evaluation of the main soundscape's descriptors across phases. Cluster analysis of personality traits identified two groups: one with higher emotional stability and openness, and another with stronger neurotic tendencies. These groups exhibited divergent perceptual patterns, with neurotic individuals reporting greater sensitivity to anthropic and chaotic soundscapes, and emotionally stable participants showing stronger appreciation for natural ones. Overall, the findings demonstrate that occupant perception cannot be understood through single-domain analyses alone. Integrating cross-domain interactions and personality-driven variability into adaptive comfort models and occupant-centric design strategies is essential to support buildings that are both energy-efficient and responsive to human diversity.

1. Introduction

Humans spend nearly 90 % of their lives indoors [1], where they are constantly exposed to multiple and simultaneous environmental stimuli, that influenced by the indoors activities, building use and uninterrupted exposure to environmental conditions over a 24-hour cycle. Human environmental perception is shaped by human senses that can be broadly categorized into two main groups: conscious (touch and taste) and unconscious (sight, hearing and smell). Unlike conscious senses, unconscious modalities cannot always be avoided at the onset of an experience, as they depend on automatic behavioural responses [2]. These stimuli, including thermal, acoustic, visual, and air quality environmental domains, collectively contribute to the overall Indoor Environmental Quality (IEQ) and strongly influence health and well-being [3,4], occupant satisfaction [5,6], and performance [7,8,3,4,9]. Improving IEQ can therefore enhance occupants' comfort and mood while simultaneously reducing buildings' energy consumption [10,11] and carbon footprint [12]. For instance, maintaining occupant comfort by running HVAC systems continuously is often energy inefficient, highlighting the need to balance energy efficiency with individual comfort through passive strategies such as natural ventilation.

Traditional comfort models, such as Fanger's Predicted Mean Vote (PMV), have long guided building regulations and design standards by specifying fixed setpoints to be respected according to building type. However, these models not only deal with a single environmental domain at a time (such as the Fanger's model just targeting thermal variables), but these also rely on average values and the "standard person" assumption, neglecting inter-individual variability. Moreover,

* Corresponding author.
E-mail address: anna.pisello@unipg.it (A.L. Pisello).

https://doi.org/10.1016/j.buildenv.2025.114140
Received 14 October 2025; Received in revised form 2 December 2025; Accepted 14 December 2025
Available online 15 December 2025
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Table 1
Specifics of the sensors used in the three series of experiments.

Dimension	Sensor name	Parameter	Sensor accuracy
Environmental	Metron Platinum resistance thermometer (PT100)	Air Temperature [°C]	±0.15 °C
	Delta OHM Thermal hygrometer	Relative Humidity [%] Air temperature [°C]	±1.5 % ±0.1 °C
	Delta OHM Luxmeter	Illuminance [lx]	±5 %
	CO_2 sensor	CO_2 concentration [ppm]	±50 ppm (+2 %)
Physiological	Empatica E4 smartwatch	Heart Rate [Hz] EDA [μS] Perimetral skin Temp. [°C]	Output resolution 0.9 nW /digit; Sensitive area: 14 mm^2 Resolution of 1 digit ~900 pS ±0.2 °C
	Muse 2 Headband	EEG amplitude [μV] (Sampling Rate: 256 Hz)	Resolution: 12-bit

recent studies have highlighted the role of non-IEQ factors (e.g., work hours, contextual factor, and personality traits) on occupants' well-being and performance [13–15], all variables that are typically neglected in common building design practices. Consequently, numerous studies report discrepancies between predicted comfort levels and occupants' self-reported perception of IEQ [16,17]. This gap underscores the limitations of classical comfort models, which fail to account for individuals' characteristics, such as psychological and personality factors, that are increasingly recognized as central to human perception [3].

In this framework, the theory of multi-domain comfort has recently emerged to contribute to face this complexity. By considering interactions between environmental factors and leveraging technological advances such as IoT, big data, and machine learning [18], multi-domain approaches enable the development of personalized comfort models (PCMs) that could better reflect individual needs and perception in terms of perceived IEQ [8]. Nevertheless, research in this field remains fragmented, as reported by Schweiker et al. [19], with some studies reporting contradictory findings. For instance, whether acoustic stimuli significantly shape thermal perception. These inconsistencies highlight the need for a deeper understanding of cross-domain effects and their interaction with human factors.

Beyond environmental variables, psychological and physiological characteristics are increasingly recognized as determinants of IEQ perception [20,21]. Most existing studies have examined correlations between objective physiological and psychological indicators and environmental parameters. Physiological variables (e.g., heart rate, skin temperature, electrodermal activity) are commonly integrated into the development of PCMs [22], whereas neurophysiological indicators (e.g., EEG), are often employed to also address the psychological dimension, to detect cognitive responses to environmental changes [23] and to assess individual's environmental perception [24]. For instance, Pourghorban et al. [25] review a wide range of studies that apply EEG analysis to evaluate human experience under different thermal conditions. However, sensing systems required to detect brain activity patterns by monitoring neurophysiological signals (i.e., EEG) are often invasive, limiting their practical use.

Current research predominantly examines the impact of IEQ on psychological restoration, mood and mental health, rather than exploring how individual psychological traits influence environmental perception. Studies have demonstrated the effects of each environmental domain on psychological restoration, mood, and minimization of mental health problems [26–29], despite these stable individual characteristics can help in explain systematic differences in how people perceive and evaluate their environment. Indeed, personality traits assessment offers a complementary perspective to neurophysiological monitoring, providing stable personal characteristic instead of moment-to-moment cognitive or psychological states. Moreover, personality traits assessment does not require invasive sensing technologies. The "Big Five" framework, based on five orthogonal dimensions (i.e., openness, conscientiousness, extraversion, agreeableness, neuroticism) [30], is widely adopted to investigate these differences. Extroverts and agreeable individuals report higher satisfaction in open-plan offices [31], whereas highly extroverted workers may also experience increased stress under certain IEQ conditions [32]. These findings underscore how personality modulates the individuals' perception, adaptation and response to indoor environments.

Focusing on environmental exposure, the acoustics domain is particularly relevant, as sounds acts not only as a physical stimuli but also a potent psychological trigger. Stress-inducing auditory inputs can activate the autonomic nervous system, altering emotional state, including feelings of relaxation and stress, heart rate, and peripheral temperature [33,34]. Concerning indoor environments, while most research studies have focused on noise reduction in workplaces and learning environments, recent approaches emphasize perception-based acoustic design to enhance health, well-being, and overall quality of life within buildings [35]. Importantly, sound perception varies across individuals: the same auditory stimulus may induce relaxation in one person but stress in another, depending on personality profiles [36]. Thus, understanding these interactions may allow for more accurate predictions of behavioural, psychological, energy-related [37], and health outcomes in response to acoustic environments.

In this framework, research exploring the relationship between personality traits and individuals' IEQ perception remains limited, and most existing studies focus primarily on understanding human behaviour under discomfort conditions [38]. Addressing these gaps is essential for developing PCMs that account for both environmental conditions and individual variability, ultimately supporting healthier, more adaptive, and energy-efficient indoor spaces. Understanding how personal characteristic [39], specifically personality traits, shape individual perception represents a crucial first step in this direction. Such insights could inform the development of PCMs that, in turn, could be implemented by Building Management Systems (BMS) to operate adaptive building layers more effectively [40,41].

Based on the above considerations, the primary research question of this study is: Do personality-driven differences influence occupants' perception of indoor conditions, particularly thermal and acoustic, within a multi-domain comfort framework? Accordingly, this study aims to investigate the cross-modal influence of acoustic and thermal comfort on occupants' perception of the indoor environment, considering individual personality traits. Specifically, it examines how different personality profiles shape occupants' evaluation of environmental quality across multiple comfort domains. By identifying these perceptual patterns, this research aims to provide useful insights for the development of personalized, occupant-centric strategies, that could be potentially implemented through BMS to enhance comfort, well-being, and long-term adaptive design. Furthermore, these findings may guide future research toward the creation of robust PCMs and evidence-based design strategies.

2. Method

An experiment procedure was designed and conducted within a test room to capture the interaction between thermal and acoustic domains and psychological traits. Subsequently, statistical analyses were conducted to investigate potential cross-domain influences and variations in perceptual responses, with personality profiles identified through a clustering process based on trait assessments.

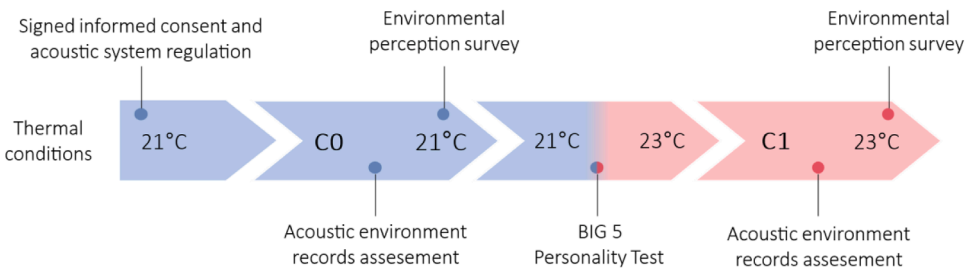


Fig. 1. Experimental procedure.

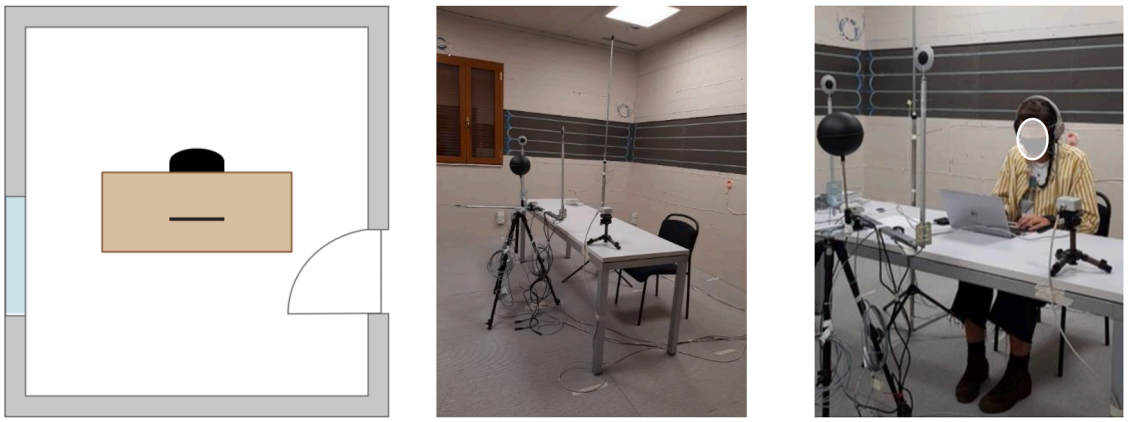


Fig. 2. Experimental environment in the NEXT.ROOM: (a) schematic of participant positioning, (b) experimental setup with the sensing system, and (c) participant during the test.

2.1. NEXT.ROOM

The NEXT.ROOM [42], a test room located in the Environmental Applied Physics Laboratory at the Engineering Campus of the University of Perugia (Italy, Cfa Köppen-Geiger climate classification [43]), is a dedicated facility for multi-domain, and highly controlled experiments. The thermal environment is regulated by two independent but complementary systems: a radiant panel system and an HVAC system, which operate concurrently to ensure spatially uniform thermal conditions. The HVAC unit, which incorporates a Controlled Mechanical Ventilation (VMC) module, allows operation at multiple fan speed settings. These settings were systematically characterized in a previous study to map and quantify the background noise levels in several positions within the NEXT.ROOM.

In addition, NEXT.ROOM is equipped with an extensive sensing system, enabling continuous monitoring and acquisition of environmental and physiological parameters throughout the entire duration of experimental sessions. The monitored variables and corresponding instrumentation are reported in Table 1. Since the NEXT.ROOM is not equipped with an integrated audio system, acoustic stimuli, during the experimental session, were provided through noise-cancelling headphones (model WH-1000XM4, by Sony).

2.2. Experimental procedure

The experimental campaign was conducted between October 2020 and January 2021. The experimental procedure (Fig. 1) consisted of four consecutive phases, beginning with an acclimation period during which participants were introduced to the indoor environment [44] and equipped with the two wearable devices: a multi-sensor wristband and noise-cancelling headphones. In this acclimation phase, lasting at least 20 min [45], the researcher outlined the study protocol using an information sheet, obtained informed consent, and assisted participants in calibrating the noise-cancelling headphones. The EEG monitoring headband was used only in the earliest sessions, as the headphones consistently interfered with signal quality and caused considerable discomfort, leading to its discontinuation in subsequent sessions. To exclude potential auditory impairments that could bias the results, a hearing test was also administered. The multi-sensor wristband, throughout the experimental session, continuously recorded physiological parameters, such as Electrodermal Activity (EDA), Heart Rate (HR) and Skin Temperature (ST), with a 1-minute sampling rate. The initial condition (C0) was carried out at a constant indoor temperature of 21 °C. Participants were invited to evaluate a total of seven different soundscapes and, subsequently, complete an environmental perception survey using a 5-point Likert scale. The questionnaire assessed perceived environmental quality, comfort and preference across environmental domains (i.e., thermal, visual, acoustic and air quality). this was followed by a 30-minute transition phase, during which the indoor temperature gradually increased to 23 °C, via the NEXT.ROOM mechanical ventilation system. During this phase, participants completed the Big Five personality questionnaire [46]. The final condition (C1) replicated the procedure of C0, with the setpoint temperature maintained at 23 °C. The experimental procedure and sequence of phases was identical for all participants, ensuring consistency across the experimental protocol.

Throughout the procedure, the participants remained alone in the NEXT.ROOM, seated beyond a N centrally positioned desk (Fig. 2). They were instructed to access an online platform via the provided laptop, which was used to administer both questionnaires and soundscapes. The soundscape order was randomized across phases and participants. However, the thermal sequence was identical for all participants, with phase C1 representing the higher temperature condition.

A total of seven soundscapes were submitted to the participants, each with distinct characteristics: (#1) Beach, (#2) Woodlands, (#3) Quiet street, (#4) Pedestrian zone, (#5) Park, (#6) Shopping mall and (#7) Busy street. Each recording had a duration of 30 s and was selected to

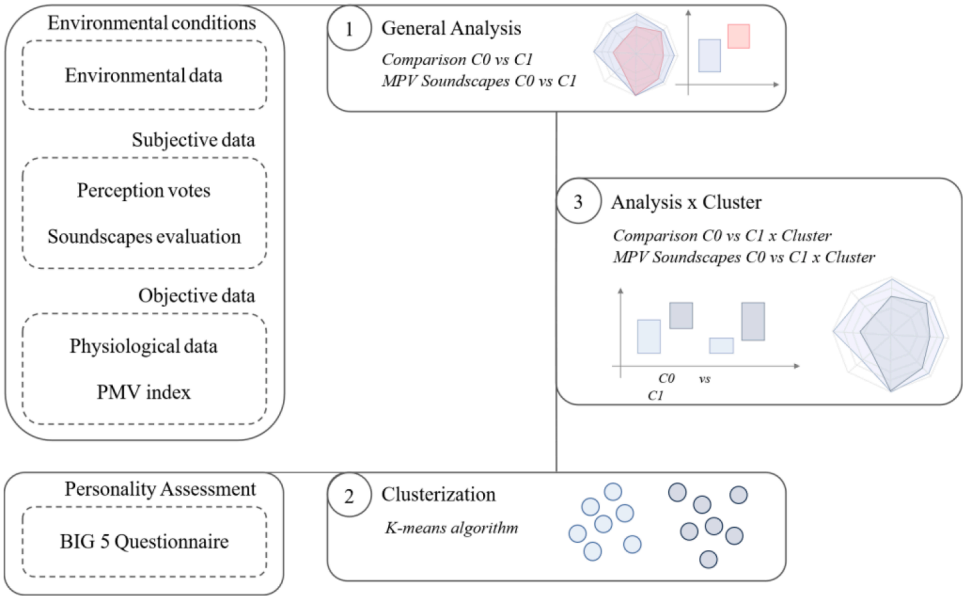


Fig. 3. Framework of the analysis conducted.

ensure a diversity of sound sources. Participants could listen to the recordings as many times as desired and were then asked to evaluate them on eight descriptors (i.e., pleasant, chaotic, lively, quiet, calm, annoying, busy and monotonous) using a 5-point Likert scale ranging from "strongly agree" to "strongly disagree".

Personality traits were assessed using the IPIP-NEO-120, a 120-items version of the BIG 5 Personality Questionnaire, which is a validated short form of the original IPIP-NEO-300 survey [46]. Participants responded on a 5-point scale, where 1 correspond to "very inaccurate" and 5 to "very accurate". Scores for each factor and facet were calculated as percentile ranks, referenced against normative data stratified by gender, country, and age. This approach allowed the classification of trait levels (i.e., high, low and average) within the relevant demographic group. Thus, the questionnaire provided a comprehensive evaluation of an individual's personality traits, offering insights into behavioural tendencies and predispositions.

2.3. Participants

Written informed consent was obtained from all participants after they were fully briefed on the aims and potential risks of the experimental protocol, which was approved by the University of Perugia's Ethics Committee. Forty volunteers were initially recruited for the experimental campaign; however, due to the complexity pf the experimental procedure and the relative data collection, 25 tests were considered as suitably of interest and reliable for the analysis. The final sample therefore consisted of 25 completed test sessions performed by 25 volunteers, recruited at the University of Perugia, including students, PhDs and Post-Docs. The sample comprised 12 males and 13 females, with an average age of 27.4 ± 2.6 years (range: 22–33 years). In order to participate in experiment, individuals were required to meet specific criteria, including being in good health, having a Body Mass Index (BMI) within the normal range, and not using drugs or consuming excess alcohol. During the experimental session, participants were required to complete a low impact listening task (metabolic rate: 1.2 MET) during C0 and C1 phases at the provided workstation. Moreover, the participants were not asked to wear specific items but were instructed to maintain throughout the experimental session a clothing insulation equal or close to 1.0 clo, typical of the heating season, as indicated by the CBE Comfort Tool [47].

2.4. Data analysis

For the statistical analysis, a sample size of 25 participants was considered, and a significance level (α) was set at 0.05. Despite the two environmental conditions, the experimental procedure yielded a unique dataset, which had been assessed for normality using Shapiro – Wilk test. Most variables included in the analysis (11 out of 15), with the exception of the environmental parameters, exhibited non-normal distributions ($p < 0.05$).

To obtain objective comfort indicators, the Predicted Mean Vote (PMV) was calculated for both thermal conditions using the "comf" package developed by M. Schweiker [48]. This package implements established models and equations for human comfort assessment. PMV indices were computed based on temperature, relative humidity, and air velocity, assuming a metabolic rate of 1.2 MET, which represents a person performing office activities, and a clothing level of 1.0 clo, indicative of autumn/winter attire. The calculations were performed in R, following the package documentation. To allow direct comparison between objective (i.e., PMV index) and subjective (i.e., participants' perception votes) comfort indicators, the five-point scale of the subjective data (−2 to +2) was linearly rescaled to match the seven-point PMV index scale (−3 to 3). This procedure preserved the relative distances between categories while aligning the ranges of the two measures for visualization purposes. Physiological signals (i.e., EDA, HR and ST) were also analysed to examine potential differences across the two thermal phases. Given the 1-minute sampling rate, the EDA signal was treated as tonic electrodermal activity (EDAt), reflecting slower, baseline variations in sympathetic arousal rather than rapid phasic responses [49,50]. For each participant, mean EDAt, HR and ST were calculated separately for C0 and C1 and compared to assess the activation of physiological regulation mechanisms associated with the thermal variation. These comparisons were tested for statistical significance using paired *t*-test, and effect size were quantified using Cohen's *d* for paired samples, computed as the mean difference divided by the standard deviation of the paired differences.

Subsequently, given the non-normality of most variables, the Spearman's rank correlation coefficients were computed to explore potential associations between subjective perceptual data and personality traits, with particular attention to each environmental domain perception and personality trait dimension. In addition, correlations among the perceptual evaluations of different domains, particularly

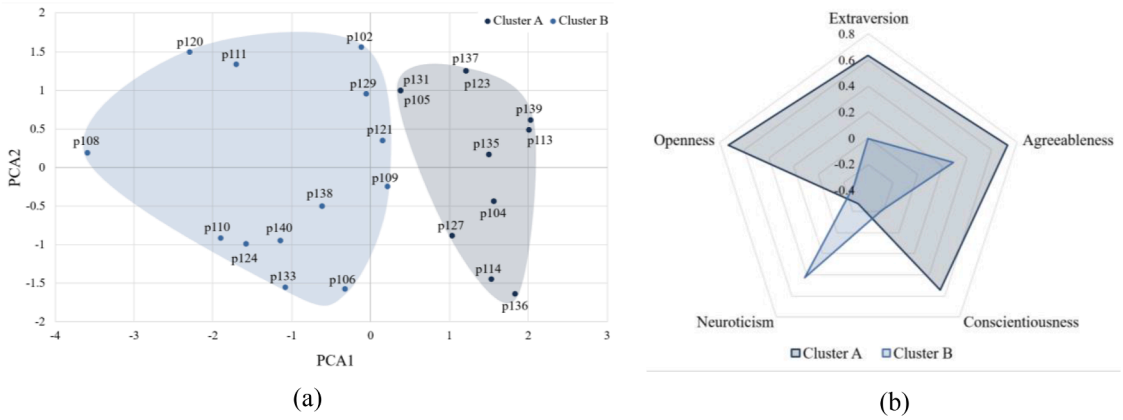


Fig. 4. a) Principal Component Analysis, with labelled points (e.g., p102, p104, p105, p106) indicating participant IDs, and b) cluster defined.

Table 2
Component loadings for each personality traits on PCA1 and PCA2.

	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness
PCA1	0.31	0.48	0.54	−0.47	0.41
PCA2	0.70	0.14	−0.41	0.45	0.35

Table 3
Correlation coefficient values between personality traits and perception votes for cluster A.

	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness
Thermal Perception Votes	0.10	−0.68	0.03	0.17	0.41
Visual Perception Votes	0.13	−0.29	0.04	0.77	0.24
Acoustic Perception Votes	0.04	−0.37	0.18	0.16	0.16
Air Quality Perception Votes	0.39	0.08	−0.15	0.08	−0.38

Table 4
Correlation coefficient values between personality traits and perception votes for cluster B.

	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness
Thermal Perception Votes	−0.33	−0.06	0.37	−0.44	−0.47
Visual Perception Votes	0.00	−0.12	0.24	−0.09	−0.46
Acoustic Perception Votes	−0.13	0.09	0.33	−0.42	0.14
Air Quality Perception Votes	−0.26	0.25	−0.07	0.42	0.56

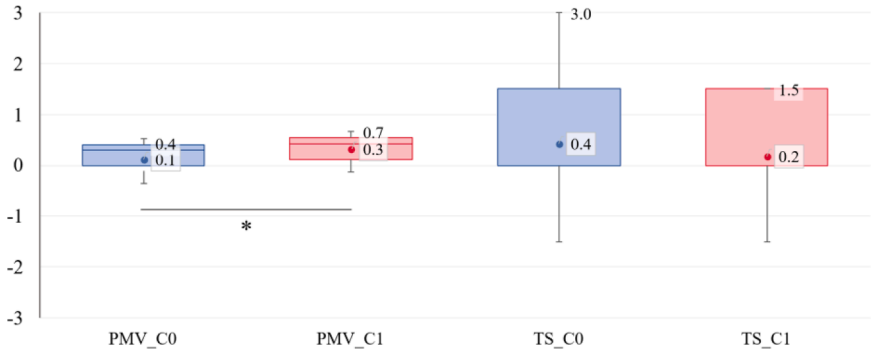


Fig. 5. PMV index and thermal sensation votes across phase C0 and C1 with "*" $p < 0.05$.

thermal and acoustic, were also analysed to investigate potential cross-domain influences.

Finally, participants' assessments of the soundscapes were further examined through a weighting scoring approach. Each response on the 5-point Likert scale was multiplied by a numeric value from 0 to 4. The rating "strongly disagree" was multiplied times 0, the rating "somewhat disagree" was multiplied by 1, the "neither agree nor disagree" rating was multiplied times 2, the "somewhat agree" rating was multiplied by 3, and the "strongly agree" rating was multiplied times 4. The mean pondered vote was then calculated ad a composite descriptor of soundscape evaluation, providing a weighted representation of participants' overall preferences. The framework of the analysis performed is

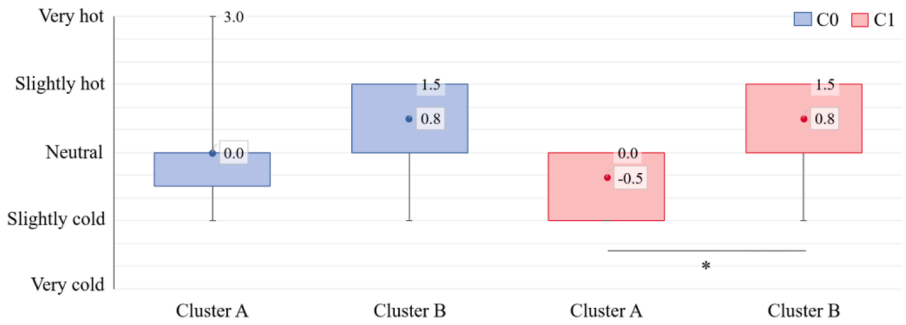


Fig. 6. Thermal sensation votes for phase C0 and C1 divided by clusters with "*" $p < 0.05$.

Table 5
Delta values between C0 and C1 for each descriptor and each soundscape.

Soundscapes	Pleasant	Chaotic	Vibrant	Uneventful	Calm	Annoying	Eventful	Monotonous
#1	0.08	0.15	0.54	−0.15	0.00	0.00	0.69	−0.62
#2	−0.07	0.13	0.07	0.20	−0.27	0.07	−0.33	0.07
#3	0.29	−0.19	0.33	0.14	0.19	0.29	0.10	−0.14
#4	−0.05	0.00	−0.05	−0.20	0.4	−0.20	0.00	0.10
#5	0.11	−0.05	−0.05	0.32	0.21	0.05	−0.16	0.11
#6	0.63*	−0.58*	0.05	0.16	0.26	−0.63*	−0.05	−0.16
#7	0.44*	−0.19	−0.31	0.69*	0.38	0.06	0.25	−0.13

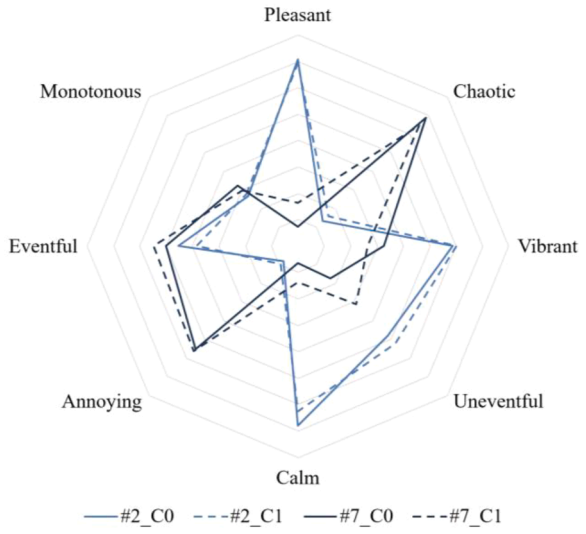


Fig. 7. Evaluation of soundscapes #2 and #7 across C0 and C1 phases.

reported in Fig. 3.

2.4.1. Clusterization

To identify common personality profiles, participants were grouped into clusters based on their personality trait assessments, aiming to capture potential patterns in perceptual responses. Each trait, measured at three levels (i.e. low, average or high) using the administered personality questionnaire, was considered equally in the clustering procedure. The K-means algorithm was applied to partition participants into cluster based on trait similarity. Each data point was assigned to the cluster whose centroid, the mean of points within that cluster, was nearest.

To determine the optimal number of clusters, the Silhouette score was computed. This metric evaluates clustering quality by considering cohesion within clusters and separation between clusters. The Silhouette score (i) ranges from −1 to 1, with values near 0 indicating points on the boundary between clusters. An average Silhouette across all points provides an overall measure of clustering quality, where higher values indicate more cohesive and well-separated cluster.

The validity of the silhouette-based solution was further corroborated using the Elbow method, a heuristic that examines the proportion of variance explained as a function of the number of clusters. The "elbow" point of the curve indicates the optimal cluster number.

Finally, Principal Component Analysis (PCA) was employed to visualize the resulting clusters in a two-dimensional representation. PCA reduces the dimensionality of the dataset while retaining the greatest variance, enabling a clear graphical representation. Component loadings indicate the contribution of each personality trait to the principal components, reflecting both the weight and direction of each variable in defining the reduced dimensions.

3. Results and discussion

3.1. Overview of environmental data

During each test, environmental parameters from the four main domains were continuously monitored at 1-minute intervals. Noise level, illuminance and CO_2 concentration remained substantially constant and were analysed to provide a reliable characterization of the surrounding environmental conditions. Regarding the visual and acoustic domains, except during the evaluation of soundscapes, the variables were considered stable, in line with the measurements reported by Vittori et al. [42] following the construction of NEXT.ROOM. Background noise was 39.7 dB in both phases C0 and C1, corresponding to a medium fan speed setting of the mechanical ventilation system, and increased to 58.1 dB during the transitional phase due to the higher fan speed required to raise the temperature. Illuminance levels were 474.2 ± 14.7 lx and 478.7 ± 5.3 lx in C0 and C1, respectively. Meanwhile, the CO_2 concentrations were 540.2 ± 46.1 ppm and 496.0 ± 26.1 ppm in C0 and C1, respectively.

The analysis focused on the thermal domain, characterized by temperature and relative humidity. In terms of temperature, the operative temperature was computed as the arithmetic mean of air temperature and mean radiant temperature, consistent with established approaches for moderate environmental conditions.

Despite the applied temperature ramp during the transition phase, the target of 23 °C was not consistently achieved. Average operative

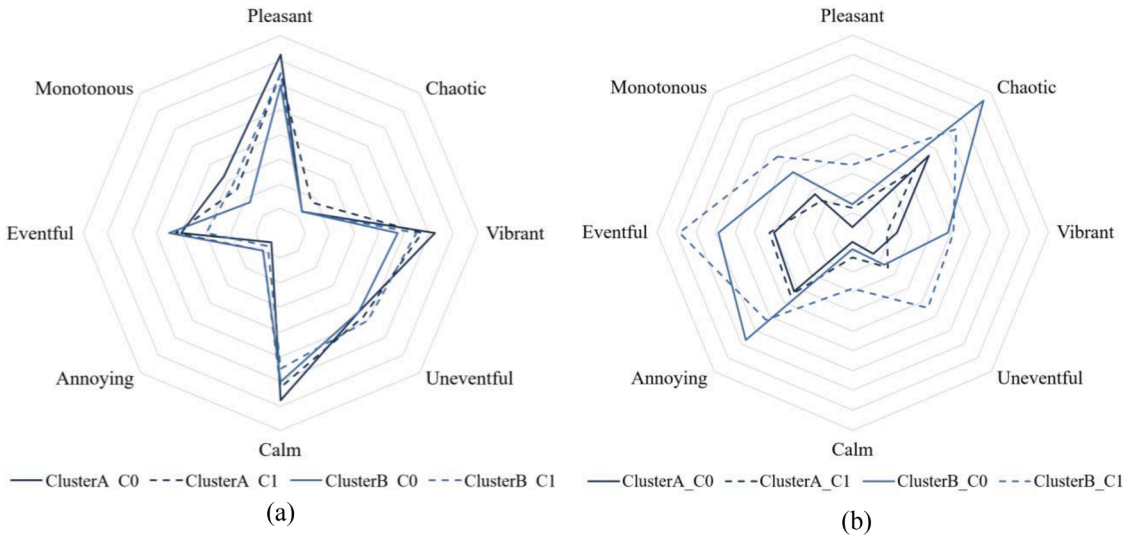


Fig. 8. Evaluation across phase C0 and C1 of a) soundscape #2 and b) soundscape #7 divided by clusters.

temperatures were found to be 21.6 ± 0.8 °C and 22.8 ± 1.0 °C in C0 and C1, respectively, with a mean variation equal to 1.22 ± 0.53 °C. Regarding relative humidity, the values recorded were 56.0 ± 9.0 % and 52.5 ± 8.2 % in C0 and C1, respectively; the small discrepancy may be attributable to the absence of control over this parameter, given that no humidifier was integrated into the HVAC system

3.2. Sample clusterization

A comparative analysis of the clustering metrics was performed to determine the appropriate number of clusters. Using the silhouette method, the two-cluster solution was identified as the most appropriate (highest coefficient = 0.26). A complementary evaluation with the elbow method suggested that two to four clusters could be suitable for the dataset, with inertia values of 85.5, 75.2 and 63.6 for 2, 3 and 4 clusters, respectively. To minimize heterogeneity in terms of cluster size and gender distribution, a two-cluster solution was adopted. Consequently, the sample segmentation was performed using the K-means machine learning algorithm, based on the five personality traits rated as low (−1), medium (0) or high (1). This resulted in a nearly even distribution: 15 participants in Cluster A (7 females and 8 males) and 10 in cluster B (6 females and 4 males).

Subsequently, using the Principal Component Analysis (PCA), the two-dimensional representation of the cluster was provided, as illustrated in Fig. 4.a. The first two Principal Components (PCs) described 64.5 % of the variance in the original data, ensuring an adequate description of the total sample. PCA1 primarily reflected self-control and emotional stability, while PCA2 captures a dimension of extraversion and emotional instability, in contradistinction to conscientiousness. The component loadings for each personality trait on PCA1 and PCA2 are reported in Table 2.

The two clusters that were identified were given the designations Cluster A and Cluster B. More specifically, Cluster A has values of 1.33 and 0.12 for PCA1 and PCA2, respectively. Conversely, Cluster B exhibited values of −1.05 and −0.10 for PCA1 and PCA2, respectively. This finding is further corroborated by the radar chart (Fig. 4.b), which demonstrates that Cluster A individuals exhibit higher levels of self-control, emotional stability, and openness to new experiences and social interactions. Conversely, Cluster B subjects exhibit pronounced tendencies towards neuroticism, emotional volatility and reduced conscientiousness.

3.3. Influence of personality traits on indoor environmental quality perception

Following the identification of two distinct clusters, the correlation between the five personality traits, assessed using the Big Five questionnaire, and participants' ratings of thermal, visual, acoustic, and air quality sensations was examined. To capture changes over time, the difference between the ratings in the initial phase (C0) and those in the final phase (C1) was calculated, producing a delta value for each comfort domain. These delta values were subsequently correlated with each personality trait to uncover potential relationships.

The findings delineate distinct personality profiles across the two clusters, underscoring their relevant role in shaping environmental perception. As shown in Table 3, Cluster A is characterized by higher levels of self-control and emotional stability. Within this group, thermal perception was negatively associated with Agreeableness and positively associated with Openness. In addition, visual perception demonstrated a significant positive association with Neuroticism. This pattern suggests that, even among individuals who are generally characterized by emotional stability, visual perception remains meaningfully linked to neurotic traits, even when such traits are expressed at very low levels.

Cluster B, by contrast, is defined by higher emotional instability, reduced self-control, and a greater prevalence of Neuroticism. For this group, Neuroticism and Openness emerged as the most influential traits across the assessed comfort domains, with the largest deviations from zero. In contrast to the findings observed in Cluster A, which included a robust positive correlation between neurosis and visual perception, however, no specific results were evident in Cluster B. These findings, detailed in Table 4, emphasize the differential role of personality traits in shaping the multisensory perception of interior environments.

These correlation values are partially aligned with findings from previous studies by Marzaban et al. [31], Hartog et al. [51], and Kallio et al. [32], which identified extraversion, agreeableness, and openness to new experiences as the personality traits most strongly associated with individuals' perceptions of their surrounding environment.

3.4. Crossed effect between thermal and acoustic domains

Regarding the crossed effect of different environmental domains, the attention in this study was focused on the thermal and acoustic domains, as these were varied during the experiment. Initially, an analysis was conducted by varying one environmental variable at a time and evaluating its impact on perceptual responses related to a different domain.

The present study examined the potential impact of the applied temperature variation on the perception and evaluation of soundscapes. Firstly, an analysis of the thermal perception expressed by participants as the temperature varied was conducted.

Furthermore, the two clusters that had been previously developed were considered in order to determine the possible influence of personality profiles on the perception of the thermal and acoustic domains.

3.4.1. Thermal perception: the role of personality profiles

According to achieved changes in NEXT.ROOM thermal environment throughout the test, observed PMV was, on average, equal to 0.11 ± 0.23 and 0.32 ± 0.27 in phases C0 and C1, respectively. This discrepancy between the two phases, although minimal, is statistically significant, with a p-value lower than 0.05. Considering the physiological response to environmental changes (i.e., +1.22 °C in temperature and −2.25 % in relative humidity from C0 to C1), the applied thermal variation can be considered as sufficient to stimulate physiological mechanisms. Previous studies have demonstrated positive correlations between temperature changes and fluctuations in physiological indicators [52,53]. In the present study, skin temperature (ST) increased by 0.76 °C ($p < 0.00$, $d = 1.9$), whereas both tonic electrodermal activity (EDAt) and heart rate (HR) decreased, respectively −0.25 μS ($p < 0.00$, $d = 0.9$) and −3.17 bpm ($p < 0.00$, $d = -2.0$). These large effect sizes indicate robust physiological responses across participants. Conversely to the findings of Heal et al. [54], who reported a positive correlation between HR and temperature in children, with increase of 8.7 bpm to 13.7 bpm per 1 °C raise, the present results showed a different cardiovascular response (i.e., decrease in bpm) under the temperature variations tested. This divergent pattern may be attributable to differences in participant age compared with those examined by Heal et al. [54], and, more importantly, to the fact that participants in the current study remained seated throughout the entire experimental protocol, likely contributing to a reduced metabolic rate.

In contrast, the analysis of participants' subjective ratings in terms of thermal sensation, as illustrated in Fig. 5, reveals a different pattern when compared to the PMV index. In fact, the thermal sensation ratings remained mostly similar between the two configurations, with average values equal to 0.40 ± 0.79 and 0.20 ± 0.73 in C0 and C1, respectively. As shown in the box plot, the mean values of the two groups are almost identical, and the distribution is highly similar, with no statistically significant difference.

The subsequent division of the total sample into the two identified clusters revealed that the different personality profiles demonstrated varied responses to changes in thermal conditions. In general, cluster B reports a warmer sensation than cluster A, but the thermal sensation expressed by cluster B remains almost constant between the two phases. Concurrently, cluster A shows a neutral stance during the initial phase C0, with a perception progressing towards negative values (i.e., cold) in phase C1.

As demonstrated in Fig. 6, the disparate responses exhibited by the two clusters in phase C1 are found to be statistically significant, with a p-value that is less than 0.05. In line with the findings of Kallio et al. [32], who reported that more extroverted individuals are more sensitive to the environmental stimuli, Cluster A displayed a different mean thermal sensation compared to Cluster B.

3.4.2. Influence of thermal discomfort on soundscapes evaluation

The impact of introducing a mild thermal discomfort between the two experimental phases (C0 and C1) on participants' soundscape evaluations was investigated. The analysis encompassed all the soundscapes submitted for evaluation by the participants. Subsequently, the variation in the evaluation of each soundscape descriptor was considered, and it was determined whether this difference was statistically significant or not. As the Table 5 illustrates, greater variation values emerged for soundscapes #1 and #2, which, according to their designations, appear to be predominantly characterized by natural sounds, and for soundscapes #6 and #7, which appear to be predominantly characterized by anthropogenic sounds.

Subsequently, the analysis focused on the soundscapes that were identified by the participants as being predominantly characterised by natural sounds (i.e. soundscape #2) and predominantly characterised by anthropogenic sounds (i.e. soundscape #7). As demonstrated in the table above, an evaluation of the soundscapes reveals a marginal decline in the primary descriptors of each audio track, with an enhancement in the assessment of non-characteristic descriptors. Regarding soundscape #2, a decrease is evident in the descriptors "Calm" and "Eventful", while a minimal variation is observed in "Pleasant". As regards soundscape #7, a marked decline was evident in the evaluation of the descriptor "Chaotic", alongside an increase in non-characteristic descriptors such as ''Pleasant'' and ''Calm''.

Here the effect of applying a temperature variation on the evaluation of the different soundscapes was analysed. No consideration was given to either contextual or personal variables. Consequently, the individuals' personality characteristics were included in the investigation and, therefore, the two defined personality profiles. As demonstrated in the preceding section, acoustic perception is influenced by different personality traits in the two clusters. Indeed, subsequent to the segmentation of the participants into the two previously identified clusters, the evaluations of the soundscapes in the two thermal phases exhibited distinct behaviours, aligned with the personality profiles. This observation is particularly evident in the two soundscapes that were the primary focus of the study (i.e., #2 and #7). In contrast, for the other soundscapes, some differences in the evaluations of the descriptors were observed, although these were not consistent in all cases.

As illustrated in the initial radar chart (Fig. 8a), significant variations are evident across four out of the eight evaluated descriptors for soundscape #2 (Woodlands). Contrasting trends are evident in the ratings assigned to pleasant, vibrant, and monotonous descriptors across the two phases. Specifically, cluster A exhibits a decline in ratings for C1, while cluster B demonstrates an increase. Despite these contradictory trends, both clusters demonstrate a convergence of ratings in C1. With regard to the eventful descriptor, cluster A demonstrates negligible change between phases, whilst cluster B exhibits a marked decrease, despite commencing from a rating analogous to cluster A in C0.

As illustrated in the second radar chart (Fig. 8b), the two clusters diverge significantly across nearly all descriptors for soundscape #7 (Busy Street). Nevertheless, they demonstrate analogous patterns for the soundscape's most characteristic descriptors (i.e. chaotic and annoying), with both clusters assigning lower ratings in C1. In contrast, for the non-characteristic descriptor (i.e., pleasant), both clusters assign higher ratings in C1, with an average difference in ratings of 0.50 and 1.00 for cluster A and B, respectively. For the remaining descriptors, contrary behaviours are observed in most cases. In particular, the ratings assigned to the vibrant, uneventful, calm and eventful descriptors by cluster A remain relatively stable. By contrast, cluster B's ratings increase in C1, except for the vibrant descriptor, which decreases. It is evident that the rating assigned to monotonous descriptor cluster A diminishes, whilst that allocated to cluster B experiences an augmentation.

These integrative findings indicate that the general analysis may not fully capture participants' evaluations across thermal phases, as it overlooks individual differences in personality. Participants in Cluster B, characterized by lower emotional stability and a heightened tendency toward neurotic traits, appear more sensitive to soundscapes dominated by anthropic sounds, which they tend to perceive as more chaotic and annoying. This effect is evident in the evaluation of the "Chaotic" attribute for soundscape #7, where Cluster B reported substantially higher ratings than Cluster A, with mean differences of −1.96 and −1.46 in C0 and C1, respectively. As shown by several previous study, the neuroticism traits have a strong positive correlation with noise environment [55], and people higher in neurotic have a greater sensitivity to noise environment [56].Conversely, Cluster A, defined by greater openness to experience and higher emotional stability, was more

responsive to soundscapes dominated by natural sounds. For Soundscape #2, their ratings exceeded those of Cluster B by–0.38 for both C0 and C1 on the "Calm" descriptor, and –0.75 and –0.13 for C0 and C1, respectively, on the "Vibrant" descriptor. Similar trends emerged for the other soundscapes: Cluster A provided higher ratings only for #1 and #2, the two stimuli devoid of anthropic elements.

In addition, participants assessed their acoustic perception of the ambient environment at the end of both thermal phases. At this stage, they were not exposed to soundscapes and wore noise-cancelling headphones; the only audible source was the HVAC system fan operating at a constant medium level. Ratings were based on a scale negative values denoted a sensation of noise and positive values denoted a sensation of silence. The two clusters displayed distinct perceptual patterns: Cluster A consistently reported a stronger perception of scores were -0.57 ± 0.94 and -0.50 ± 1.02 in C0 and C1, respectively, compared with -1.36 ± 1.03 and -1.73 ± 0.47 in C0 and C1, respectively, for Cluster A.

4. Conclusion

In light of the growing need to design interior spaces that ensure the occupants well-being while minimizing energy consumption, research on environmental comfort is increasingly shifting toward an individual-centred approach. Within this context, emerging studies focus on the development of personalised models aimed at predicting users' perceptions and satisfaction with the built environment [57]. It is well established that each individual exhibits diverse physiological, psychological and preferential responses, shaped by the interaction between environmental conditions, contextual factors, personal characteristics and past experiences [58]. These differences substantially influence both perception and behaviour, suggesting that unexpected or suboptimal occupant actions, often unaccounted for during the design or simulation phase, can lead to increased energy consumption. The findings of this study may provide a basis for the formulation of strategies to mitigate discomfort caused by environmental factors through the implementation of post-occupancy strategies, such as Personal Environmental Comfort Systems (PECS) to create more tailored and occupant-centric environments. Furthermore, considering the diverse stakeholders involved in the building sector, the insights gained here may inform the operation and coordination of different building adaptive layers. In particular, in open space offices or co-working spaces, understanding how personality traits influence occupants' perception of IEQ can support the design of environments, enhancing both individual comfort and energy consumption.

In this context, the present study examined the cross-modal effects of two environmental domains – thermal and acoustic – and the moderating role of personality characteristics on the perception of IEQ. This exploratory investigation constitutes a component of a more extensive research programme that is oriented towards the utilisation of the user as the pivotal focal point within the design and management processes of constructed environments.

To address these questions, an experiment was conducted in which environmental conditions were controlled, specifically through variations in temperature and exposure to soundscapes comprising natural and anthropogenic components. This methodology allowed the investigation of cross-domain interactions, specifically whether certain thermal conditions influence the perception of soundscapes, potentially the salience of noisier auditory stimuli. Conversely, the study also investigated whether certain soundscapes could attenuate the perception of temperature variations, potentially reducing the likelihood of occupant actions such as adjusting thermostats or opening windows.

Firstly, the analysis of environmental conditions confirmed that the applied temperature variation, although minimal, was sufficient to generate a statistically significant difference in the PMV index. Moreover, the analysis of the physiological signals (i.e., EDAt, ST, HR) reported the activation of regulatory mechanisms, aimed to maintain equilibrium with the surrounding environment, confirming the physiological response to the applied temperature variation. However, this temperature variation was not mirrored in the participants' thermal sensation ratings, which remained constant between the two phases (i.e., C0 and C1). Considered together with the soundscapes evaluations, these results provide further evidence of the potential reciprocal influence between thermal and acoustic domains. Specifically, for the soundscapes under consideration (i.e., #2 – Woodlands and #7 – Busy street), participants reported lower ratings for the main descriptors, suggesting that minor thermal fluctuations can attenuate the perception of both pleasant and unpleasant auditory stimuli, and vice versa, that auditory stimuli can mitigate the perception of thermal variations.

Subsequently, two personality profiles were identified on the basis of the five characteristic traits assessed by the Big Five Questionnaire. Cluster A, marked by higher levels of self-control, emotional stability and openness, showed greater sensitivity to natural sounds, consistently rating them as calm and pleasant, while perceiving no substantial differences in anthropogenic sounds across thermal phases. Conversely, Cluster B, characterized by neurotic tendencies, lower emotional stability, and reduced self-control, was more reactive to anthropogenic sounds, which they evaluated as annoying and chaotic. Interestingly, under higher temperature conditions, the Cluster B reported a partial reduction of negative evaluations, with corresponding increases in calmness and pleasantness.

Taken together, these findings highlight both the cross-domain interaction between thermal and acoustic comfort and the mediating role of personality traits in shaping individual environmental perception. Even when divided into personality-based clusters, the results consistently demonstrated a potential reciprocal influence between the two domains, though with distinct perceptual patterns across profiles.

In conclusion, individual characteristics significantly influence the perception of environmental domains, shaping different ways of experiencing and adapting to indoor spaces. Nevertheless, the study is subject to several limitations. Firstly, the modest sample size, which precluded the development of probabilistic models of perception and comfort, based on the environmental conditions experienced. The modest sample size also limited the clustering analysis, resulting in a low silhouette score and affecting the robustness of the clustering outcome. Moreover, the experiment was conducted in a test room, a controlled setting without contextual factors, and used noise-cancelling headphones, which may have created an artificial scenario in the participants' acoustic domain. Future research should extend these investigations to real environments, such as Living Labs, to assess cross-domain effects in real contexts. Furthermore, targeted use of soundscapes at specific frequencies, delivered via both headphones and room-based speaker, may hold potential for enhancing cognitive performance and concentration while mitigating the perceived impact of environmental fluctuations. Finally, integrating self-reported environmental preferences of individuals [59], together with additional physiological measures such as EEG, could yield a more comprehensive understanding of cognitive and affective processes underlying multi-domain perception (Fig. 7).

CRediT authorship contribution statement

Agnese Chiucchiù: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Veronica Martins Gnecco:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Ilaria Pigliautile:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation. **Anna Laura Pisello:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the European Union's Horizon Europe Research and Innovation Actions programme grant SONATA – Situation aware OrchestratioN of AdapTive Architecture (G.A. no. 101137507). Funded by the European Union. Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.The authors' acknowledgments are also due to the Italian Ministry of Research for supporting the young researcher PRIN project WePOP (Prot.2022RKLB3J). Anna Laura Pisello also thank the European Union - NextGenerationEU - for supporting the Italian Ministry of University and Research (MUR) National Innovation Ecosystem grant ECS00000041 - VITALITY - CUP I33C2200133000.

Data availability

Data will be made available on request.

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